How-To Library

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# Creating Match Rule Sets in Provisioning Tool

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# Abstract

You can use the Provisioning tool to create match rule sets based on a sample data and the match training. This article describes how to create match rule sets using the Provisioning tool.

# Supported Versions

- Multidomian MDM 10.4

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# Match Rule Sets Overview

A match rule set is a logical collection of match rules. Match rules determine whether a record matches with another record. You can create match rule sets in the Hub Console or the Provisioning tool. You can use the Hub Console to manually create the match rule sets.

You can use the Provisioning tool to automatically create match rule sets based on a sample data and the match training. The match engine matches the records in the sample data set and identifies 10 pairs of records for you to review. Based on your review, the match engine creates the match rules and provides a confidence score for the match rules. If the confidence score is low, retrain the match engine until you get a high match confidence score. Use the match rules to run the match process.

The quality of the match rules depends on the data set that you use. Ensure that the data set contains a combination of matching records, partially matching records, and unique records to create accurate match rules.

**Note:** Informatica recommends that you create match rule sets in non-production environments only.

# Scenario

Your organization has customer information stored in a database that might contain duplicate records for many customers. You want to create match rules to match duplicate customer records. You are not equipped to use the Hub Console to create match rule sets. With the help of sample data set and Provisioning tool, you create match rule sets that you can use to run the match process.

# Configuring the Hub Server Properties

Before you create match rule sets, ensure that you configure the Hub Server properties related to match training in the `cmxserver.properties` file.

Configure the following properties for match training:

**cmx.match.training.confidence.threshold**

Optional. Minimum match confidence score required to create a match rule set in the Provisioning tool. Default is 85.

**cmx.match.training.data.encoding**

Optional. Configures encoding for match training in the Provisioning tool. Set to 1 to enable encoding for match training. Default is 0.

Ensure that you use the same value that the `cmx.server.match.server_encoding` property uses.

For more information about the Hub Server properties, see the *Multidomain MDM Configuration Guide*.

# Prerequisites for Creating Match Rule Sets

You can create a match rule set for a business entity based on a sample data set.

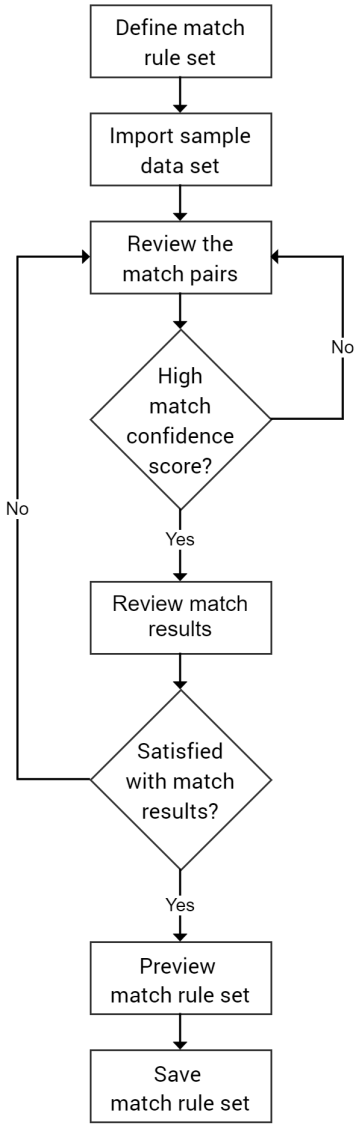
Before you create a match rule set for a business entity, verify that your environment meets the following prerequisites:

- No match rule sets exist for the business entity.
- No match rule sets exist for the root base object of the business entity.
- No records exist for the root base object of the business entity.
- The data set contains a minimum of 1000 records and a maximum of 10,000 records.
- Your user role has permission to import bulk data. For more information about assigning permissions, see the *Multidomain MDM Security Guide*.

# Creating Match Rule Sets

Creating match rule sets using the Provisioning tool is easier and faster. Before you create a rule set, ensure that you have created an optimal data set.

The following flow chart shows the process of creating a match rule set:




## *Step 1. Define a Match Rule Set*

When you define a data set, you must specify the data population. A data population contains intelligence about name, address, and other identification information that is typical for a given population. Use an appropriate data population to improve the match accuracy.

1. In the Provisioning tool, click **Configuration** > **Match Rule Sets**.
2. Click **New Match Rule Set**.

   The **New Match Rule Set** page appears.
3. Enter the name of the match rule set.

4. Optionally, enter a description for the match rule set.
5. Select a business entity for which no match rule set exists.
6. Select a data population.

## *Step 2. Import Sample Data Set*

Upload a data set that contains a minimum of 1000 records and a maximum of 10,000 records. The match engine automatically maps the fields in the sample data with the existing business entity fields.

1. Under **Match Training Data Set**, click **Upload**.

   The **Match Training Data Set Import** dialog box appears.
2. To upload a file, drag a CSV or Excel file to the **Upload File** box, or click **Browse** and select a file.

   The match training data set preview and the file properties appear.

   **Note:** Click **Change File** if you want to change the sample data set.
3. In the **File Properties** panel, specify the following file properties:

| Property | Description |
|---|---|
| Delimiter | A character that represents the break between data values in the import file. |
| Text Qualifier | Symbol used in the file to enclose a string. |
| Code Page | Character encoding used in the file. Character encoding is the mapping of the characters from a language or group of languages to hexadecimal code. |
| Import Data From Line | Line from where you want to import data. |
| Has column headers | Indicates whether the uploaded file includes column headers. |
| Column Headers in Row | Number of the row that serves as the column header in the file. |
| Regional settings | Select the check box to update the following settings:<br>- Date Pattern: Date format for the date fields in the import file.<br>- Decimal Separator: Symbol used to separate decimal part of a number.<br>- Thousand Separator: Symbol used to separate groups of thousands. |

4. Click **Next**.

   The match engine maps the column headers in the uploaded file with the business entity fields.
5. To update any of the auto-generated field mapping, drag the source column headers to the relevant target business entity fields.
6. Click **Save**.

   A confirmation message to delete any existing records in the root base object that the selected business entity uses appears.
7. To continue, click **Yes**.

   A message to confirm the delete operation appears.
8. Click **Yes** to continue with match training.

   The **Match Training** page appears.

## *Step 3. Train the Match Engine*

Review the data pairs. After you review all the data pairs, the match engine displays the match results based on the match rule set that identified the match pairs. The match engine requires at least eight confirmed review inputs to display the match results.

1. On the **Match Training** page, select a key type to use for key generation and click **Continue**.

   The first data pair appears for review.
2. Review the data pair, and click one of the following options:
   - **Match**. If the records are same.
   - **Not Sure**. If you are not sure whether the records are same.
   - **Not a Match**. If the records are not same.
3. After you complete reviewing 10 data pairs, click **Show Match Results**.

   The match confidence meter and the match results appear.

## *Step 4. Preview and Save the Match Rule Set*

Review the match results and match confidence score and decide whether to continue training, or save the match rule set.

A low match confidence score indicates that the prediction of the match engine did not match with your inputs for the match pairs. You must continue the match training until you get a high confidence score.

A high match confidence score indicates that the prediction of the match engine almost matched with your inputs for the match pairs. To be sure, check the match results. If you are satisfied with the match results, proceed to preview and save the match rule set.

1. If the match confidence score is low, click **Continue Training**.
2. If the match confidence score is high, do the following tasks:
   a. To review the match results, select a training session from the list of match trainings.
   b. To view the details of a match pair, click the match score.
   c. Select the best training session, and click **Next**.
   d. Review the match rules, and click **Save**.

      The match rule set appears in the **Match Rule Sets** page.

# Author

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